

Tanish Saroj

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EDUCATION

Theem College of Engineering

Bachelor of Engineering in Computer Science | GPA: 8.0+ / 10

Mumbai, India

Expected May 2026

ACHIEVEMENTS

Winner, Smart India Hackathon (SIH) 2025 – 1st Rank Nationwide | Problem Statement #25164

Designed autonomous drone defense system for Bharat Electronics Limited (BEL) using swarm AI

PROJECTS

Insight – AI Data Analyst Agent | Python, LangChain, Streamlit, Groq API, Llama 3 | [Live Demo](#)

- Built conversational LLM agent that processes natural language queries against CSV datasets – executes sandboxed Python, generates vintage newspaper-styled visualisations, exports formatted Word reports
- Engineered aesthetic consistency by injecting chart styling directives into the agent's system prompt, ensuring Matplotlib output matches the UI palette across all LLM-generated visualisations
- Integrated Groq API for ultra-low latency inference; supports Llama 3 70B, Mistral, and Compound-Beta with configurable temperature and model switching via sidebar

Heimdall – AI Code Review Agent | Python, FastAPI, LangChain, Groq API, Llama 3 70B | [GitHub](#)

- Built LLM-powered agent triggered by GitHub webhook events – fetches commit diffs, routes through Llama 3 70B for analysis, and posts structured reviews (bugs, tests, docs) as commit comments
- Designed prompt templates for structured output: summary, issue classification, documentation drafts, and test case suggestions per commit

Persona-Adaptive QA Bot | Python, PyTorch, QLoRA, Hugging Face, Phi-3

- Fine-tuned Microsoft Phi-3-mini (3.8B) with QLoRA and PEFT to dynamically adapt response style by user persona – Student vs. Expert; achieved 67% style adaptation, 52% factual accuracy on held-out sets
- Optimised training pipeline for free Colab T4 via 4-bit quantisation – full fine-tune at zero cost

Chronicle – Prompt Version Control Platform | Python, FastAPI, PostgreSQL, React | Collaborative

- Designed architecture for LLM prompt management system: immutable versioning, production promotion model, alias history audit trail, and variable injection engine with strict mode
- Architected LLM evaluation pipeline with exact-match, LLM-judge, and confidence-calibration evaluators; results surfaced to teams via plain-English deployment recommendations

SimAstra – SIH 2025 National Winner | Python, PyBullet, Swarm Algorithms, NumPy

- Engineered decentralised Hunter-Killer swarm algorithm for autonomous drone threat neutralisation without central controller; validated across 100+ combat scenarios in PyBullet simulation

TECHNICAL SKILLS

Languages: Python, SQL, C++

LLM & Agents: LangChain, Groq API, Llama 3, Mistral, Phi-3, Hugging Face Transformers, MCP, Prompt Engineering

ML / Training: PyTorch, QLoRA, PEFT, 4-bit Quantisation, Fine-tuning, LightGBM, Scikit-learn

Backend & Infra: FastAPI, Docker, PostgreSQL, SQLAlchemy, Git, Streamlit, python-docx

Libraries: Pandas, NumPy, Matplotlib, Seaborn, PyBullet